

Instructor Marking Assist : An AI-Powered Grading Assistant

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The Problem & Solution Overview

- ◉ *The Challenge:* Heavy administrative workloads delay student feedback.
- ◉ *The Solution:* A Human-Centered AI (HCAI) prototype for grading "heavy lifting".
- ◉ *The Core Goal:* Instructors maintain full control over every final decision.

Research Questions

- ⦿ How can AI align with SALT's specific grading rubrics?
- ⦿ How does the interface support "Human-in-the-Loop" overrides?
- ⦿ How do we ensure data privacy and ethical standards?

ML/AI Model

- ◉ *Step 1:* OCR and NLP to process submissions.
- ◉ *Step 2:* Instructor settings would adjust how grading thresholds are interpreted.
- ◉ *Step 3:* A Large Language Model (LLM) would compare answers to the rubric.
- ◉ *Step 4:* Output would include a grade, feedback, and reasoning summary.

Evaluation & Testing

- ◉ Historical grading data would be used as the baseline for evaluation.
- ◉ Agreement with instructor scoring would be a core validation metric.
- ◉ Robustness testing would focus on file quality and formatting variation.
- ◉ Output consistency would be tested across different grading rigour settings.
- ◉ User acceptance testing would measure trust in grades and explanations.

Human-Centered AI Principles

- ⦿ *Empowerment*: Enhances the instructor's role without replacing it
- ⦿ *Transparency*: Eliminates the "black box" with clear grade justifications.
- ⦿ *Inclusion*: Features Dark Mode and adjustable text sizes for accessibility

Ethical Guidelines & Data Privacy

- ◉ Student data is processed locally within sessions to ensure privacy
- ◉ Rigorous adherence to Microsoft's Responsible AI Standards.
- ◉ Rigour settings and custom rubrics help mitigate algorithmic bias

Conclusion

- The interface was designed to support efficient and transparent evaluation.
- Instructor oversight was treated as a required part of the system.
- Any future implementation would depend on trust, fairness, and privacy.
- The proposed system supports teaching by saving time, not replacing judgment.

AI Usage Statement

This project leveraged Large Language Models to help create the HTML/CSS prototype and polish technical documentation. All final design choices, ethical guidelines, and grading logic were carefully reviewed and verified by humans to ensure academic integrity and full alignment with SAIT standards.

References

- Microsoft. (2022). Microsoft Responsible AI Standard, v2.0.
<https://query.prod.cms.rt.microsoft.com/cms/api/am/binary/RE5cmFI>
- Southern Alberta Institute of Technology. (2026). ARTI-406-A: Human-Centered AI - Project: AI-powered application [Course assignment description and rubric]. Brightspace. <https://learn.sait.ca>
- Southern Alberta Institute of Technology. (n.d.). Brand visual identity: Colour.
<https://www.sait.ca/brand/visual-identity/colour>